

A Practical Guide to Mathematical optimization

TOPIC -- MATHEMATICAL OPTIMIZATION

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Mechanics Problems in rigid body dynamics (in particular articulated rigid body dynamics) often require mathematical programming techniques, since you can view rigid body dynamics as attempting to solve an ordinary differential equation on a constraint manifold; the constraints are various nonlinear geometric constraints such as "these two points must always coincide", "this surface must not penetrate any other", or "this point must always lie somewhere on this curve". Also, the problem of computing contact forces can be done by solving a linear complementarity problem, which can also be viewed as a QP (quadratic programming) problem. Many design problems can also be expressed as optimization programs. This application is called design optimization. One subset is the engineering optimization, and another recent and growing subset of this field is multidisciplinary design optimization, which, while useful in many problems, has in particular been applied to aerospace engineering problems. This approach may be applied in cosmology and astrophysics.

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Given: a function $f : A \rightarrow \mathbb{R}$ from some set A to the real numbers Sought: an element $x_0 \in A$ such that $f(x_0) \leq f(x)$ for all $x \in A$ ("minimization") or such that $f(x_0) \geq f(x)$ for all $x \in A$ ("maximization"). Such a formulation is called an optimization problem or a mathematical programming problem (a term not directly related to computer programming, but still in use for example in linear programming – see History below). Many real-world and theoretical problems may be modeled in this general framework. Since the following is valid:

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argmax $x, y \cos x y$, subject to: $x \in [-5, 5], y \in \mathbb{R}$,
 $\underset{\{x,y\}}{\operatorname{arg\,max}} x \cos y, \text{subject to: } x \in [-5,5], y \in \mathbb{R}, \}$

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asks for the maximum value of the objective function $2x$, where x may be any real number. In this case, there is no such maximum as the objective function is unbounded, so the answer is "infinity" or "undefined".

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Global convergence More generally, if the objective function is not a quadratic function, then many optimization methods use other methods to ensure that some subsequence of iterations converges to an optimal solution. The first and still popular method for ensuring convergence relies on line searches, which optimize a function along one dimension. A second and increasingly popular method for ensuring convergence uses trust regions. Both line searches and trust regions are used in modern methods of non-differentiable optimization. Usually, a global optimizer is much slower than advanced local optimizers (such as BFGS), so often an efficient global optimizer can be constructed by starting the local optimizer from different starting points.

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Besides (finitely terminating) algorithms and (convergent) iterative methods, there are heuristics. A heuristic is any algorithm which is not guaranteed (mathematically) to find the solution, but which is nevertheless useful in certain practical situations. List of some well-known heuristics:

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Feasibility problem The satisfiability problem, also called the feasibility problem, is just the problem of finding any feasible solution at all without regard to objective value. This can be regarded as the special case of mathematical optimization where the objective value is the same for every solution, and thus any solution is optimal. Many optimization algorithms need to start from a feasible point. One way to obtain such a point is to relax the feasibility conditions using a slack variable; with enough slack, any starting point is feasible. Then, minimize that slack variable until the slack is null or negative.

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Sensitivity and continuity of optima The envelope theorem describes how the value of an optimal solution changes when an underlying parameter changes. The process of computing this change is called comparative statics. The maximum theorem of Claude Berge (1963) describes the continuity of an optimal solution as a function of underlying parameters.

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Necessary conditions for optimality One of Fermat's theorems states that optima of unconstrained problems are found at stationary points, where the first derivative or the gradient of the objective function is zero (see first derivative test). More generally, they may be found at critical points, where the first

derivative or gradient of the objective function is zero or is undefined, or on the boundary of the choice set. An equation (or set of equations) stating that the first derivative(s) equal(s) zero at an interior optimum is called a 'first-order condition' or a set of first-order conditions. Optima of equality-constrained problems can be found by the Lagrange multiplier method. The optima of problems with equality and/or inequality constraints can be found using the 'Karush–Kuhn–Tucker conditions'.

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Mathematical optimization (alternatively spelled optimisation) or mathematical programming is the selection of a best element, with regard to some criteria, from some set of available alternatives. It is generally divided into two subfields: discrete optimization and continuous optimization. Optimization problems arise in all quantitative disciplines from computer science and engineering to operations research and economics, and the development of solution methods has been of interest in mathematics for centuries. In the more general approach, an optimization problem consists of maximizing or minimizing a real function by systematically choosing input values from within an allowed set and computing the value of the function. The generalization of optimization theory and techniques to other formulations constitutes a large area of applied mathematics.

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Major subfields Convex programming studies the case when the objective function is convex (minimization) or concave (maximization) and the constraint set is convex. This can be viewed as a particular case of nonlinear programming or as generalization of linear or convex quadratic programming. Linear programming (LP), a type of convex programming, studies the case in which the objective function f is linear and the constraints are specified using only linear equalities and inequalities. Such a constraint set is called a polyhedron or a polytope if it is bounded. Second-order cone programming (SOCP) is a convex program, and includes certain types of quadratic programs. Semidefinite programming (SDP) is a subfield of convex optimization where the underlying variables are semidefinite matrices. It is a generalization of linear and convex quadratic programming. Conic programming is a general form of convex programming. LP, SOCP and SDP can all be viewed as conic programs with the appropriate type of cone. Geometric programming is a technique whereby objective and inequality constraints expressed as posynomials and equality constraints as monomials can be transformed into a convex program. Integer programming studies linear programs in which some or all variables are constrained to take on integer values. This is not convex, and in general much

more difficult than regular linear programming. Quadratic programming allows the objective function to have quadratic terms, while the feasible set must be specified with linear equalities and inequalities. For specific forms of the quadratic term, this is a type of convex programming. Fractional programming studies optimization of ratios of two nonlinear functions. The special class of concave fractional programs can be transformed to a convex optimization problem. Nonlinear programming studies the general case in which the objective function or the constraints or both contain nonlinear parts. This may or may not be a convex program. In general, whether the program is convex affects the difficulty of solving it. Stochastic programming studies the case in which some of the constraints or parameters depend on random variables. Robust optimization is, like stochastic programming, an attempt to capture uncertainty in the data underlying the optimization problem. Robust optimization aims to find solutions that are valid under all possible realizations of the uncertainties defined by an uncertainty set. Combinatorial optimization is concerned with problems where the set of feasible solutions is discrete or can be reduced to a discrete one. Stochastic optimization is used with random (noisy) function measurements or random inputs in the search process. Infinite-dimensional optimization studies the case when the set of feasible solutions is a subset of an infinite-dimensional space, such as a space of functions. Heuristics and metaheuristics make few or no assumptions about the problem being optimized. Usually, heuristics do not guarantee that any optimal solution need be found. On the other hand, heuristics are used to find approximate solutions for many complicated optimization problems. Constraint satisfaction studies the case in which the objective function f is constant (this is used in artificial intelligence, particularly in automated reasoning). Constraint programming is a programming paradigm wherein relations between variables are stated in the form of constraints. Disjunctive programming is used where at least one constraint must be satisfied but not all. It is of particular use in scheduling. Space mapping is a concept for modeling and optimization of an engineering system to high-fidelity (fine) model accuracy exploiting a suitable physically meaningful coarse or surrogate model. In a number of subfields, the techniques are designed primarily for optimization in dynamic contexts (that is, decision making over time):